10.4 Presenting Data

Lesson Introduction

Benchmarks	MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data. MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range. MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	Learning Target	• I can explain data using multiple representations.

	Building On	Working Toward
Benchmark Coherence	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Essential Questions	• How can you accurately and truthfully display data in different ways? • How can you present and explain data in a clear way? • Why is it helpful to represent data using visual displays like box plots and histograms?
Lesson Narrative	This lesson can be considered the second part of the previous lesson in which students surveyed the class to collect data. Students will continue to work with their group members to finalize their data displays and present their findings to the class.

Component Summary	10.4.1 Warm-Up (~5-10 min.)	Work with group members to finalize presentations *(SE p. 220)*	10.4.2 Activity (~30 min.)	Present information from surveys to the class *(SE p. 220)*
	10.4.3 Wrap-Up (~5 min.)	Students reflect on what they have learned from the presentations *(SE p. 221)*		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Histogram	• Anchor chart • Chart markers • (Optional) Projector • (Optional) Computer	• Most of this lesson is student presentations. Consider creating a presentation framework for students to follow while they are presenting.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may incorrectly display their data.
Students may incorrectly read the histogram they create.</td><td>If students are digitally displaying data displays, consider having the documents ready before the lesson on the projector/SMART board.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Group Presentations
Students work with small groups to present their findings from the surveys they created in the previous lesson. To ensure students are actively participating throughout the presentations, consider having students synthesize a group’s findings in their own words. Alternatively, encourage students to have questions prepared to ask the groups during and after the presentation.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
MTR 1 is embedded throughout this lesson, as students work with group members to effectively analyze and clearly present information they have collected during the previous lesson. Encourage students who are not in the presenting group to analyze the information being presented and ask clarifying questions to deepen understanding. This ensures all students are staying engaged with the content even when not presenting.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">This lesson provides multiple entry points for students to engage with the content through group presentations. Students present their findings using multiple data displays, including histograms, frequency tables, and box plots. These displays offer multiple visual entry points for students. If students struggle to understand the data presented in the box plot, for example, encourage them to use one of the other representations to make sense of the data. Learners are offered auditory entry points through the oral presentations as well.

Additional differentiation opportunities are denoted throughout the lesson guide.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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10.4.1 Warm-Up: Presentation Preparation

Component Overview: *Students will finalize their presentations with their groups. Circulate as students work and ensure groups have a clear plan for how each member will contribute to the presentation.*

10.4.1 Warm-Up: Presentation Preparation

1. Meet with your group from the prior lesson to finalize your data displays for the group presentations.
   Answers will vary.

Consider setting a timer for this activity to push students to stick to a schedule. While students are finalizing their presentations, briefly review the groups' work to ensure appropriateness and a good spread of data.

If data displays were finalized during the prior lesson, consider allowing students to find comparative data points for the survey data they collected. For example, if one of the survey questions was "How many movies have you seen in the past year?", have students research how many movies on average people in America watched in the past year to present with data they collected.

10.4.2 Activity: Presenting Data

Component Overview: *Students will present their data as a group to the class. Consider choosing group order ahead of time and displaying a timer so students know how long they have to present. Groups will discuss their findings from the data they collected. Encourage students to write down feedback or question answers as they are listening to other groups speak and allow time for Q & A after groups have presented their findings.*

10.4.2 Activity: Presenting Data

Assign a group member to ask each question below for the presentation.

1. What conclusion did you reach about the class?
2. Do the displays clearly demonstrate this conclusion?
3. Are there any unusual features in the data?
4. Is there anything else that the group noticed from the data displays?
5. Explain which type of graph, box plot or histogram, was more useful to display the data from your group survey.
6. Explain if the data collected is truthful. If the data was not truthful, explain what could have been done to ensure the data collected from the survey was truthful.
7. Is there anything else that the group wonders about the results?

   Answers will vary.

Consider assigning questions to each student or electing a group leader to assign the questions. Alternatively, have students present their information interview style, where one group member will ask the questions and the others will take turns answering. Students could also present their answers to the questions without the questions being asked. Encourage students to take notes on the other presentations.

Consider providing students with a printout of a presentation or a personal version of an anchor chart to better see what is being presented.

10.4.3 Wrap-Up: Cool Down

Component Overview: *Students reflect on and explain the advantages and disadvantages of using graphs to present data.*

10.4.3 Wrap-Up: Cool Down

1. What is an advantage of using graphs to present findings?
   Answers will vary. Having a visual representation is helpful for explaining the shape of a distribution and identifying features like outliers and clusters.

2. Are there disadvantages of using graphs to present findings? Explain.
   Answers will vary. It takes more time to make or find accurate graphs.

3. What new information did you learn about the class today from the different survey results?
   Answers will vary.

For students who might struggle representing their conceptual understanding through writing, consider alternative methods to explain thinking. Students may record their responses using an app or present them orally to the teacher or a partner.